

Over-Squashing vs. Adversarial Robustness in GNNs

An “Add vs. Remove Edge” Investigation

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The question

Two GNN problems that look unrelated, but both are about **structural sensitivity**:

- **Over-squashing** — long-range signal squeezed through bottlenecks and lost. Output *not sensitive enough* to distant nodes.
- **Adversarial fragility** — a few edited edges flip the prediction. Output *too sensitive* to malicious structural change.

Do methods that fix over-squashing also help robustness?

The spine: add vs. remove edge

- Over-squashing rewiring: FoSR **adds** edges; spectral pruning **removes** edges (both raise the spectral gap).
- Adversarial attack: **adds** dissimilar edges. Defense (GCN-Jaccard): **removes** them.

Hypothesis: *adding* to fix over-squashing may *hurt* robustness (more edges to exploit); *removing* for over-squashing may *transfer* to robustness (removes like a defense).

Research questions

- RQ2 (main)** Do over-squashing fixes (add-based FoSR, remove-based spectral pruning) improve adversarial robustness — vs. an undefended GCN and vs. a purpose-built defense — and at what cost to clean accuracy?
- RQ1** Does the attack worsen the over-squashing *itself*? (perturbation vs. long-range information loss)
- RQ3 (stretch)** Can a single measure (Jacobian sensitivity) capture *both* over-squashing and adversarial fragility?

We lead with **RQ2**; RQ1 comes almost for free from the same probe; RQ3 is a stretch.

Related work & positioning

- **The building blocks already exist** — over-squashing rewiring (FoSR, spectral pruning), the attack (Metattack), the defense (GCN-Jaccard). We introduce *no* new method.
 - **The link is understudied** — to our knowledge, rewiring/over-squashing surveys do not discuss adversarial robustness, and the robustness literature does not engage over-squashing.
- ⇒ **Our contribution:** a controlled *add-vs-remove* study connecting the two lines of work, reporting only real, reproducible numbers.

Closest related: a rewiring method (GraphTorque) improves robustness — but via edge *removal*, consistent with our finding.

Methods (the design)

Method	Targets	Edge op	Role
GCN (vanilla)	—	none	baseline
FoSR	over-squashing	add	“add” arm
ProxyDelete	over-squashing	remove	“remove” arm
GCN-Jaccard	defense	remove	defense reference
DropEdge	regularization	random	random-removal control

FoSR vs. ProxyDelete = direction; ProxyDelete vs. Jaccard = criterion.

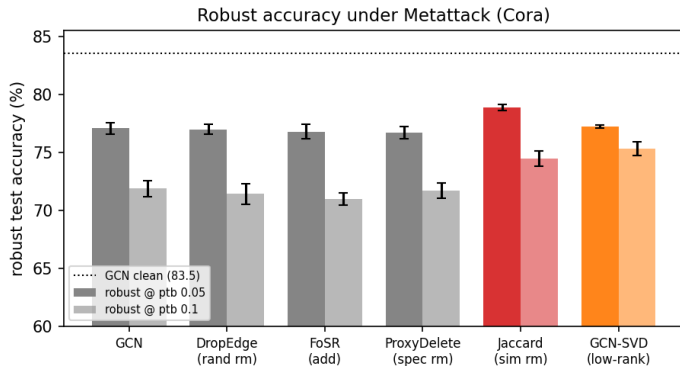
Setup

- **Cora & Citeseer** node classification, DeepRobust `nettask` split, CPU.
- Attack: Metattack (poisoning), `ptb` {0.05, 0.10}; cached once, reused.
- Pipeline: `clean` $\rightarrow$ Metattack $\rightarrow$ method $\rightarrow$ train GCN $\rightarrow$ eval.
- Metric: `clean` / robust accuracy, robustness **gap** = `clean` – robust.

Dataset	Nodes*	Classes	Features
Cora	2,485	7	1,433
Citeseer	2,110	6	3,703

*standard `nettask` split (largest connected component)

Results: add vs. remove under attack (Cora)



Four arms (GCN, DropEdge, FoSR, ProxyDelete) cluster at the GCN level; **the two purpose-built defenses separate** — Jaccard (similarity) & SVD (low-rank). The add/remove *direction* does not matter — the editing *criterion* does.

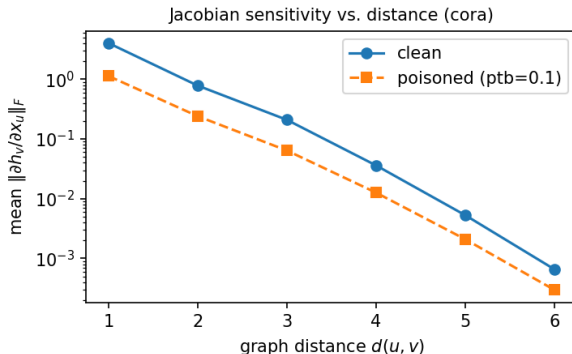
Why does Jaccard win? (mechanism)

Metattack almost purely **adds** edges (503 added, 3 removed at ptb 0.10). How many does each method delete?

Method (ptb .10)	# removed	attack edges rm.	recall
GCN-Jaccard	797	250	0.50
ProxyDelete (spectral)	643	93	0.19
DropEdge (random)	643	59	0.12
FoSR (add)	0	0	0.00

Chance recall ≈ 0.12 . Jaccard deletes **half** the attack edges ($\sim 4\times$ chance) $\Rightarrow$ it undoes the attack.
Spectral/random $\approx$ chance; add removes none.

Over-squashing probe (RQ1)



Sensitivity decays $\sim 5\times/\text{hop} \Rightarrow$ over-squashing is real. Poisoning shifts the level but **not** the decay rate $\Rightarrow$ over-squashing & fragility are decoupled (matches RQ2).

Adaptive attacker: evadable but self-defeating

A Jaccard-aware attacker adds edges between feature-**similar** nodes (the filter won't remove them).

- Evasion works at the filter: Jaccard removes **0%** of these edges.
 - But the attack turns weak: GCN drops only 0.7/2.1 pts (vs 6.5/11.7 for Metattack).
- ⇒ Edges that fool the model (dissimilar) are the ones Jaccard removes; edges that evade Jaccard (similar) barely fool it — **potency & detectability are coupled**.

Heuristic (not gradient-optimal) adaptive attack; optimal adaptive = future work.

Findings: which knob decides robustness?

Ablation summary: which “knob” decides adversarial robustness?

Factor we varied	Held fixed	Result	What we learned
Add vs. remove (FoSR vs. ProxyDelete)	over-squashing goal	both fail to defend	direction is not what matters
Edit criterion (spectral / similarity / low-rank)	that we edit edges at all	only attack-aware editing defends (Jaccard, SVD)	the criterion is the key
Edge budget (253 vs. 643 edges)	the method	spectral removal still fails	it is not a budget artifact
Dataset (Cora vs. Citeseer)	the method	same ranking of methods	the finding generalizes
Attacker (non-adaptive vs. adaptive)	the defense	evades the filter but turns weak	potency & detectability are coupled
Over-squashing measure (clean vs. poisoned graph)	the model	decay rate unchanged	over-squashing & fragility are decoupled

Conclusion

- Over-squashing mitigations (FoSR add, ProxyDelete spectral-remove) **do not** improve adversarial robustness on Cora / Metattack.
- Only attack-aware defenses defend — Jaccard (similarity) & SVD (low-rank); robustness comes from targeting the attacker's edges, not from optimizing the spectral gap.
- Honest negative result: the over-squashing $\rightarrow$ robustness bridge is *not* supported here.
- Robust to budget (spectral removal still fails at 643 edges) & replicated on **Cora + Citeseer**. A defense-aware attacker evades Jaccard only by turning weak.
- Next: gradient-optimal adaptive attack, RQ3 (Jacobian $\rightarrow$ fragility).

Thank you

Questions?